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Clinical course of Merkel cell carcinoma: A DeCOG multicenter study of 1049 patients

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ABSTRACT

Background: Merkel cell carcinoma (MCC) is a highly aggressive skin cancer with neuroendocrine differentiation characterized by frequent recurrences. Large epidemiological databases (e.g., SEER, IARC) lack granularity in analyzing associations between tumor, patient characteristics, locoregional interventions and recurrence patterns.

Patients and methods: Within the pre-immunotherapy era (1998–2017) the DeCOG MCC registry included 1049 patients with histopathologically confirmed MCC. Patient/tumor characteristics, treatment details, and outcomes were analyzed. Primary endpoints were progression-free probability (PFP) and disease-specific survival (DSS).

Results: Median age at diagnosis was 74 years; 50.4 % were males. Primary tumors most frequently occurred on the head/neck (32.2 %) and upper extremities (29.1 %). One-third of patients presented with stage ≥IIIA disease. At a median follow-up of 10 years, 36- and 60-months PFP rates were 69.0 % and 63.9 %, respectively; DSS rates were 86.9 % and 82.6 %. Surgical margins of 1–2 cm provided the best PFP and DSS improvement; margins > 2 cm did not further improve clinical outcome. Similarly, for stage IIIA patients a complete lymph node dissection (CLND) did neither improve PFP nor DSS. Early radiotherapy (<8 weeks post-diagnosis) significantly improved PFP (HR 1.36) and DSS (HR 1.79). Expansion of radiotherapy to lymph node bed showed no additional

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benefit. Patients with multiple metastases at first recurrence had poorer DSS (HR 2.0) compared to those with single metastases, irrespective of locoregional or distant spread.
Conclusions: MCC outcomes are optimized with surgical margins of 1–2 cm and timely adjuvant radiotherapy. Larger margins, CLND in stage IIIA, or extended treatment radiation fields did not improve survival outcomes.

Introduction

Merkel cell carcinoma (MCC) is a rare skin cancer characterized by neuroendocrine differentiation.[1] Its carcinogenesis is based either on the integration of the Merkel cell polyomavirus or ultraviolet (UV) mutagenesis. Several countries have demonstrated an increasing incidence of MCC over time among non-Hispanic Whites using population-based cancer registries.[2,3] This trend is likely due to aging populations and the higher incidence of MCC among the elderly.

While valuable population-based data regarding incidence and survival can only be reliably obtained from cancer registries covering large populations, population-based cancer registries primarily assess the probability of survival of MCC patients at the population level[2]. Therefore, these registries are neither optimized for determining progression free probability (PFP) due to incomplete reporting of recurrence or progression, nor for the impact of different initial locoregional therapies on disease progression, as therapeutic measures are not recorded in the required level of detail.

Initial treatment of MCC typically involves a combination of therapeutic measures, depending on stage and extent of disease.[4–7] Surgical excision is the standard treatment for localized MCC tumors. Wide local excision with safety margins of up to 3 cm combined with sentinel lymph node biopsy (SLNB) is recommended to remove the primary tumor and to check if the cancer had spread to draining lymph nodes. Indeed, SLNB is an important part of the disease staging process since the American Joint Committee on Cancer (AJCC) v8 classification. It is subdivided in a clinical staging for patients who had no SLNB performed, and a pathological staging which is considering SLNB results.[8] Complete lymph node dissection (CLND) is often recommended for patients with a positive SLNB. Radiation therapy targeting the tumor bed and the draining lymph node region is frequently used for adjuvant treatment after initial surgery. However, the value of irradiating the lymph node region has recently been questioned.[9]

After successful initial treatment, MCC recurs in up to 40 % of patients after initial locoregional treatment, with 90 % of these recurrences arising within the first 3 years after diagnosis.[10] Given this high risk of recurrence, MCC's immunogenicity and the efficacy of immunotherapy in advanced disease, several adjuvant immunotherapy trials are ongoing and the first promising interim results have been reported.[11]

Several factors influencing the risk of recurrence in MCC patients are well established, such as site of the primary tumor, stage at first diagnosis, sex, age, and immunosuppression.[10,12] However, due to the rarity of MCC and the lack of epidemiological registries with detailed clinical data particularly on the locoregional therapy, there remains uncertainty about the impact of different types of initial locoregional treatment on the further course of MCC disease. The absence of locoregional therapy-specific recurrence data poses a significant challenge in designing appropriate treatment and surveillance plans for individual patients based on their recurrence risk. Furthermore, this lack of data complicates the development of adjuvant and neoadjuvant therapies, as recently became obvious by our experience in the ADMEC-O trial investigating adjuvant nivolumab immunotherapy versus observation.[11] Thus, accurately estimating case numbers for adjuvant and neoadjuvant trials becomes exceedingly difficult, hampering effective statistical planning.[13]

The evaluation of specialized disease registries provides a more detailed characterization of patient and tumor characteristics, a more precise description of therapeutic measures used, and a better follow-up of the course of disease in terms of progression or recurrence. Therefore, for the present study we utilized the database of our large national multicenter Dermatological Cooperative Oncology Group (DeCOG) registry of histologically confirmed, prospectively enrolled MCC patients to evaluate not only the established prognostic risk factors, but also the impact of different types of locoregional therapies on progression-free probability (PFP) and disease-specific survival (DSS). It is important to note that we deliberately limited the analysis period to the time before the widespread use of ICIs for treating advanced MCC.[14,15] This approach was used to reflect the spontaneous course of MCC disease, meaning to investigate patient, tumor, and locoregional treatment characteristics impacting a DSS that was not yet influenced by effective systemic therapies. To ensure that our patient cohort is truly representative, we compared the demographic data from our disease-specific clinical registry with the corresponding data from the epidemiological cancer registry of North Rhine-Westphalia (Germany), which covers a population of 18 million people, in order to exclude possible biases, e.g. due to the predominant reporting from referral centers of the former.[16] Additionally, we performed an alignment of our results with previously established risk factors for disease recurrence.

This study investigated the impact of patient, tumor, and locoregional treatment characteristics on survival outcomes of MCC patients. Our findings demonstrate that the impact of the currently recommended interventions on disease progression often falls short of expectations, underscoring the need to for a critical reassessment of existing treatment guidelines.

Patients and methods

Between January 1998 and December 2017, 1076 patients with histopathological confirmed MCC were prospectively enrolled into the multicenter registry of the DeCOG (Supplementary Table S1). Of these, 27 patients had to be excluded due to inadequate follow-up (n = 20) or incomplete baseline characteristics (n = 7), leaving 1049 patients suitable for analysis (Supplementary Figure S1). The strict separation of identity data (trustee office) and medical data (registry), as well as the central identity and consent management prevented duplicate registration of patients. Data were extracted from patient records at the respective institutions and documented in an electronic case report form to be centrally merged for analysis. Clinical and tumor-specific data, such as sex, date of first diagnosis, age at first diagnosis, anatomic localization of the primary, histological subtype of the primary and its mitotic rate, primary tumor thickness and diameter, TNM stage, final extent of surgical safety margin, sentinel lymph node and CLND status, time and type of adjuvant radiotherapy, date and type of first and subsequent recurrences, location of recurrence, time, type, localization and treatment of recurrences, as well as survival status were collected.

Disease progression was defined as the first recurrence following locoregional treatment, excluding deaths unrelated to MCC, reflecting the older demographic typically affected by the disease. Disease progression in multiple locations in a single patient at the same time were classified as one event. Progression-free probability (PFP) comprised the time from first diagnosis to first recurrence. Non-MCC deaths were treated as a competing risk for progression-free probability. Disease-specific survival (DSS) was defined as time from first diagnosis to death from MCC. Overall survival (OS) was defined as time from first diagnosis to death from any cause. Follow-up was estimated using the reverse Kaplan-Meier method as time until censoring. The reverse Kaplan Meier estimates the follow-up time with reversed status indicator, so that the event of interest (progression, death from any cause) becomes the censor.[17] Patients were censored at the date of last follow-up, if they did not meet the endpoint (progression or death).

To avoid a bias by only reporting statistically significant results, and to assess the precision of our estimates, we additionally calculated and reported confidence intervals (CI).[18,19] PFP, DSS and OS were estimated with the Kaplan–Meier product limit method. All patients with specified information on covariates of interest were included. To account for competing risks in estimating PFP, the Fine-Gray model was performed.[20,21] Complete case analysis without missing values was performed. Multivariable Cox regression analysis was performed to estimate hazard ratios for possible confounders for progression-free probability. As the data are based on real-world data inverse probability treatment weighting (IPTW) was used to assess possible confounders.[22,23] IPTW was performed for age (modeled as a continuous variable), sex (female or male), and tumor stage (pI, cI, pII, cII, IIIA, IIIB, IV). IPTW is commonly employed in observational studies to balance baseline characteristics between exposed and unexposed groups. This method is particularly critical for accounting for disease stage when evaluating the impact of locoregional therapy (e.g., surgery, radiation therapy), as disease stage strongly influences the selection of therapeutic strategies.[24] Additionally, we performed multiple imputation using the assumption of missing at random to replace missing values.[25] Missing data was estimated using a polytomous regression imputation for unordered categorical data, employing recurrences, adjuvant radiotherapy, stage at first diagnosis, localization of primary, sex, histological subtype, age at first diagnosis as predictors. Crude and adjusted hazard ratio (HR) and corresponding 95 % confidence intervals (CI) were estimated using the Cox proportional hazards model. Statistical analyses were performed using SPSS Statistics software (version 27.0, International Business Machines [IBM], Armonk, NY), R (R version 3.2–11), and RStudio.[26]

The study registry was approved by the institutional ethics committee of the University of Würzburg (124/05), the institutional data protection officer, and the respective local ethics committees as necessary.

Results

Patient and tumor characteristics

Among the 1049 patients included in the analysis, 49.6 % (n = 520) were female. The median age at first diagnosis of MCC was 74.0 years (range: 28–105). Females were at diagnosis approximately two years older than males, with a median age of 75.0 years (95 % CI 68.0–81.0) compared to 73.0 years (95 % CI 66.0–79.0). Concurrent malignancies were common among the patients, with 14.7 % diagnosed with other types of skin cancers, including 131 cases of epithelial skin cancer. Furthermore, 12.9 % of patients had other non-cutaneous solid cancers, and 25 patients were affected by both skin and other cancers. Relevant immunosuppression was identified in only 4.4 % of patients, including 31 with hematologic malignancies and 15 receiving immunosuppressive medications. The most common localizations of primary tumors were the head/neck region (32.2 %, n = 338) and the upper extremities (29.1 %, n = 305), followed by the lower extremities (17.9 %, n = 188). The median horizontal tumor diameter was 1.8 cm, and the median tumor thickness was 8.6 mm. Histological subtypes of MCC were recorded in 40 % of the patients, with an equal distribution among trabecular, intermediate, and small cell types. A high mitotic rate was observed in 280 out of 368 reports (76.0 %). At initial diagnosis, the majority of patients (57.1 %, n = 599) presented with localized disease, without lymph node involvement or distant metastasis, corresponding to disease stages I and II according to AJCCv8. This finding may reflect the practice pattern, with sentinel lymph node biopsy (SLNB) used for pathological staging in 60 % of cases. Detailed baseline characteristics are provided in Table 1.

To assess potential biases associated with the predominance of data

Table 1
Baseline patient and tumor characteristics.

Patients	Total (n = 1049)	% of specified
Median age fist diagnosis (range/ IQR[a])	74.0 (28.0; 105.0/ 67.0; 80.0)	
Sex		
Female	520	49.6
Male	529	50.4
Immunosuppression		
Yes	46	4.4
No	996	94.9
Not specified	7	0.7
Prior skin cancer	154	14.7
NMSC	131	12.9
Melanoma	23	2.2
Other malignancy (solid and hematological)	132	12.6
Solid tumor	94	9.0
Haematological malignancy	31	3.0
Not clearly labeled	7	0.7
Anatomic localisation of primary		
Head/Neck	338	32.2
Trunk	75	7.1
Upper extremity	305	29.1
Lower extremity	188	17.9
Occult	39	3.7
Not specified (9.9 %)[b]	104	
Median horizontal tumor diameter in cm (range/ IQR[a]) Not specified n = 513 (49.0 %)[b]	1.8 (0.0; 29.0/ 1.0; 3.0)	
Histological subtype of primary		
Small-cell	158	38.0
Intermediate	136	32.7
Trabecular	122	29.3
Not specified (60.3 %)[b]	633	
Mitotic rate of primary		
High	280	76.1
Mediums	74	20.1
Low	14	3.8
Not specified (65 %)[b]	681	
Median tumor thickness in mm (range/ IQR[a]) Not specified n = 707 (67.5 %)[b]	8.6 (0.0; 80.0/5.0; 15.0)	
T stage at primary diagnosis		
T0	39	4.6
T1	543	64.3
T2	149	17.6
T3	39	4.6
T4	75	8.9
Not specified (19.4 %)[b]	204	
N stage at primary diagnosis		
N0	724	73.0
N1	268	27.0
N1a	131	13.2
N1b	134	13.5
N1 not specified (1.1 %)[b]	3	0.3
Not specified (5.4 %)[b]	57	
M stage at primary diagnosis		
M0	950	95.7
M1	55	4.3
Not specified (4.2 %)[b]	44	
Stage at primary diagnosis AJCC 8[th] ed.[c]		
pI	238	26.9
cI	173	19.6
pII	105	11.9
cII	83	9.4
IIIA	109	12.3
IIIB	121	13.7
IV	55	6.2

[a] IQR: Interquartile range
[b] percentages refer to patients with specified characteristics.
[c] Due to change of tumor classification 165 patients were not classified according to AJCC 8th edition.

from referral centers, we compared baseline patient and tumor characteristics in our cohort, such as sex, age at primary diagnosis, anatomical localization of the primary tumor, and stage at primary diagnosis, with data from the population-based North Rhine-Westphalia (Germany) cancer registry from a comparable time period.[16] Overall, the demographic data were highly comparable between the two datasets, supporting the robustness and generalizability of our findings (Supplementary Table S2). One obvious difference observed was a slightly younger median age at diagnosis in the DeCOG cohort (75 years for females and 73 years for males) compared with the cancer registry of North Rhine-Westphalia (79 years for females and 77 years for males; see Supplementary Table S2).

Locoregional treatment

Detailed information on locoregional management, including surgical safety margins, sentinel lymph node biopsy (SLNB), and adjuvant radiotherapy, was available for the majority of patients (Table 2). Among the 728 patients with complete data, 64.4 % (n = 469) underwent primary excision with a final safety margin greater than 1.0 cm, and SLNB was performed in 60.0 % (n = 533). When micrometastases were detected by SLNB, a CLND was subsequently performed in 59.1 % of patients (n = 91). Elective lymph node dissection was carried out in 18.3 % (n = 65). Adjuvant radiotherapy was administered to 76.0 % of patients (n = 597), with detailed information on type and timing available for 431 patients. Among these, 34.3 % (n = 148) received adjuvant radiation to the tumor bed alone, while 65.7 % (n = 283) received radiation to both the tumor bed and the draining lymph node region. Notably, in 42.5 % (n = 183) of patients, radiation therapy was applied more than 8 weeks following definitive surgery of the primary.

Disease course in the overall patient cohort

The median follow-up time for the entire patient cohort was 60.1 months (interquartile range [IQR] 12.7–72.9). The median PFP was 74.9 months (95 %CI 65.0–99.8). The progression-free probabilities at 24, 36, and 60 months were 74.0 % (95 %CI:71.1–77.0), 69.0 % (95 %CI 65.9–72.2), and 63.9 % (95 %CI 60.6–67.5), respectively. During the follow-up period, 266 deaths (25.4 %) were recorded in the cohort of 1049 patients, of which 130 (12.4 %) were attributed to MCC. Notably, 136 deaths (13.0 %) occurred prior to documented disease recurrence, representing competing risks in this cohort of elderly patients. This led to marked differences between DSS and OS. The median DSS for the entire cohort was not reached (95 %CI NR-NR), whereas the median OS was 167.0 months (95 %CI 165.0-NR). In more detail, the DSS probabilities at 24, 36 and 60 months were 90.8 % (95 %CI 88.8–92.8), 86.9 % (95 %CI 84.5–89.3) and 82.6 % (95 %CI 79.7–85.6); the respective OS probabilities were 84.5 % (95 %CI 82.0–86.9), 78.4 % (95 %CI 75.6–81.3) and 70.6 % (95 %CI 67.3–74.0).

Immunosuppressed patients, defined as those with hematologic malignancies, immunosuppressive disorders, or treatment with agents such as methotrexate, azathioprine, mycophenolate mofetil, JAK inhibitors, or calcineurin inhibitors, demonstrated an elevated risk of both, progression of and mortality from MCC. This is reflected by a HR of 1.84 (95 %CI 1.03–3.28) for PFP and 1.82 (95 %CI 0.70–4.73) for DSS in this patient group, both adjusted for age, sex, and disease stage. In contrast, patients with concurrent skin or other solid tumors did not exhibit a more aggressive disease course.

Among the 1049 patients, 322 (30.7 %; Supplementary Table S3) experienced disease progression, primarily as MCC recurrence. The availability of detailed data on time, type and localization of this initial progression allowed for a thorough analysis of its features and the subsequent clinical course. The majority of first progressions involved regional lymph nodes (61.8 %; n = 199) or satellite/in-transit metastases (24.2 %; n = 78). However, over the entire disease course, distant metastases developed in nearly half of the patients (47.8 %; n = 154), with a subset of patients (13.4 %; n = 43) presenting with both distant and locoregional disease concurrently at the time of first progression.

Patients whose first disease progression involved two or more simultaneous metastases had worse disease-specific survival compared with those presenting with only one solitary metastasis. After adjustment for age, sex, and tumor stage, the HR for DSS was 2.0 (95 %CI 1.31–3.08) for two metastases, and 2.16 (95 % CI 1.39–3.36) for three or more metastases (Supplementary Figure S2A). Notably, the patient's prognosis was similarly poor regardless of whether the simultaneous multifocal metastases were locoregional or distant (Supplementary Figure S2B). Analyses of time to first progression, stratified by the site of initial recurrence, demonstrated that nearly three-quarters of events occurred within the first two years following primary diagnosis. This was primarily driven by clinically detectable skin, lymph node, or soft-tissue metastases (Supplementary Figure S2C). Notably, organ metastases rarely presented as first recurrence within the first year.

Disease course by stage, sex, localization of the primary, and histological subtype

Our data underscore the critical role of SLNB in detailed prognostic staging. Patients classified as clinical stage I or II had a considerably higher risk of recurrence compared with those in whom micrometastases were excluded by pathological examination. The HRs for PFP were 1.00 for pathological stage I (stage pI), 1.44 for clinical stage I (stage cI), 1.77 for stage pII, and 3.18 for stage cII (Figure 1A, Figure 2). The fact that this difference reached statistical significance only for stage II, despite a higher number of stage I patients in this analysis, aligns with the assumption that lymph node micrometastases are more prevalent in stage II disease. Specifically, PFP probabilities for stage pII, cII and IIIA at 60 months were 71.1 % (95 %CI 61.2–82.5), 53.2 % (95 %CI 40.6–69.6) and 44.9 % (95 %CI 35.5–56.9); respective DSS probabilities were 87.8 % (95 %CI 80.6–95.8), 69.2 % (95 %CI 56.7–84.4) and 76.2 % (95 %CI 67.3–86.3). Thus, stage cII exhibits a prognostic profile more similar to stage IIIA than to stage pII.

Regarding established clinical risk factors, we confirmed that males had worse PFP compared to females (HR 1.47, 95 %CI:1.18–1.85). However, contrary to previous reports, we did not find that primary tumors of the head and neck region were associated with the poorest prognosis (Figure 1E and F).[27] Instead, primaries of the trunk carried the highest risk for disease progression and the poorest DSS (PFP HR 1.79, 95 %CI 1.20–2.65; DSS HR 3.40, 95 %CI 2.01–5.73). Importantly,

Table 2
Locoregional treatment.

	Total (n = 1049)	% of specified
Final safety margin		
≤ 1.0 cm	220	30.2
1.1–2.0 cm	225	30.9
2.1–3.0 cm	244	33.5
No surgery	39	5.4
Not specified (30.6 %)[d]	321	
Sentinel lymph node biopsy		
Yes	533	60.0
Positive	154	16.1
Negative	366	39.3
Not specified (2.4 %)[d]	13	
No	356	40.0
Not specified (15.3 %)[d]	160	
Adjuvant radiotherapy		
Yes	597	76.0
Tumor bed < 8 weeks	82	10.4
Tumor bed and locoregional lymph node < 8 weeks	166	21.1
Tumor bed > 8 weeks	66	8.4
Tumor bed and locoregional lymph node > 8 weeks	117	14.9
Not specified (27.8 %)[d]	166	21.1
No	189	24.3
Not specified (25.1 %)[d]	263	

[d] Percentages refer to patients with specified characteristics.

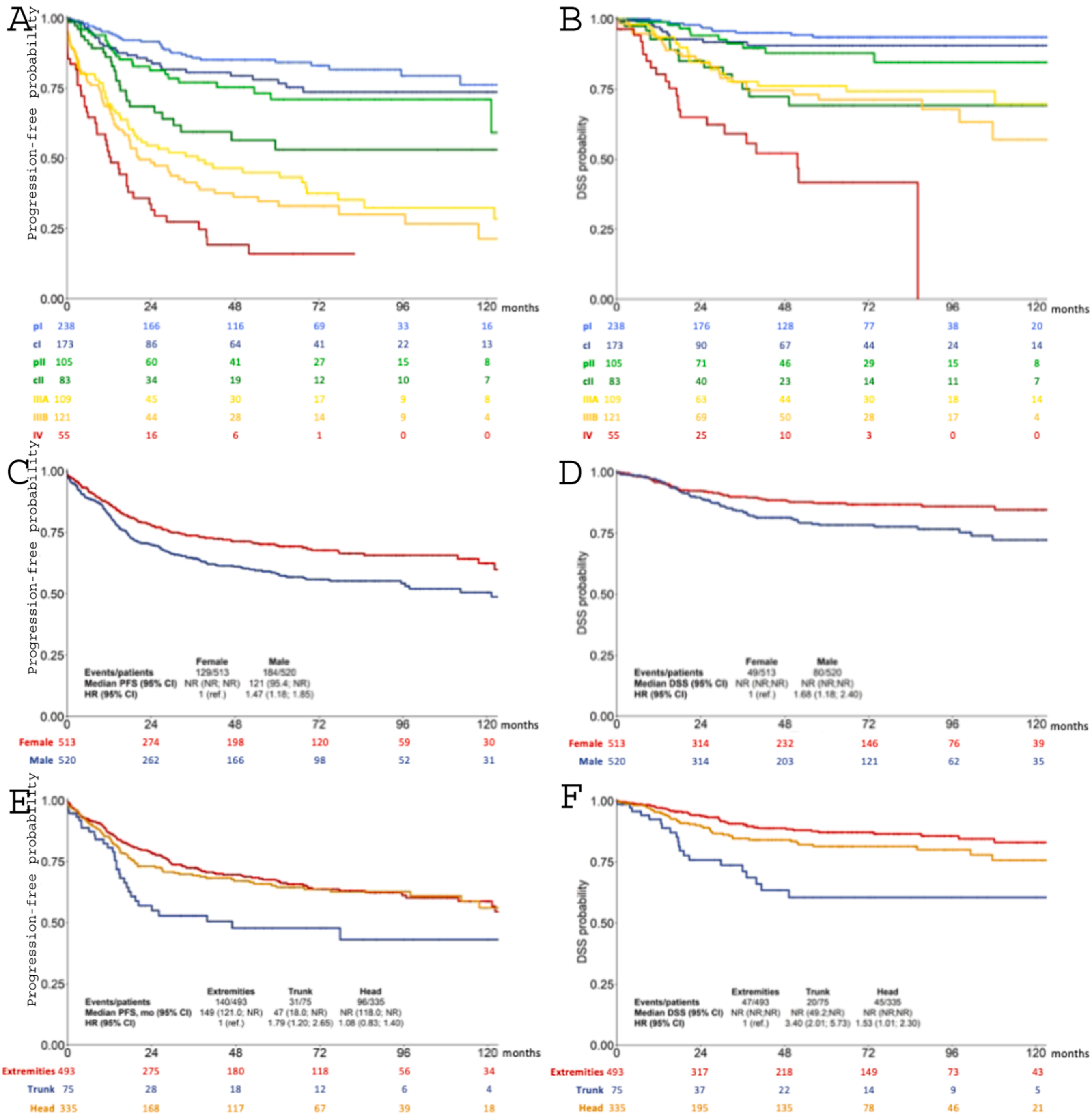

Figure 1. Progression-free probability and disease-specific survival by stage (A, B), sex (C, D), localization of primary (E, F). The Kaplan Meier estimates were calculated with unadjusted, crude data.

primary tumors on the trunk had a median horizontal diameter of 3.0 cm (mean 3.74 cm) compared to a median horizontal diameter of 2.0 cm (mean 2.38 cm) for primary tumors of the extremities, and 1.2 cm (mean 1.93) for those of the head and neck region, which may explain in part the impaired prognosis. The availability of detailed clinical data for a large number of patients further allowed us to examine the prognostic significance of the primary tumors' histopathologic type, i.e. trabecular, intermediate or small cell (total n = 414), demonstrating that the small cell type appear to carry the highest risk of recurrence (HR 1.71, 95 %CI 1.13–2.59).

Disease course by locoregional treatment

A key advantage of specialized clinical cancer registries, compared with population-based cancer registries, is their ability to provide detailed data on therapeutic management and disease progression. This high level of granularity of the DeCOG registry enabled a thorough evaluation of the impact of various locoregional therapy strategies, including the extent of the final surgical safety margin, the implementation of CLND following a positive SLNB, and the timing and extent of adjuvant radiotherapy — topics that remain subjects of ongoing discussion.

Multivariable Cox regression analyses showed that stage at first diagnosis and male sex were associated with higher risk for progression (Supplementary Table 4 and 5). To account for missing data for other parameters, we performed multiple imputation to exclude possible biases in the reporting (Supplementary Methods). The influence of

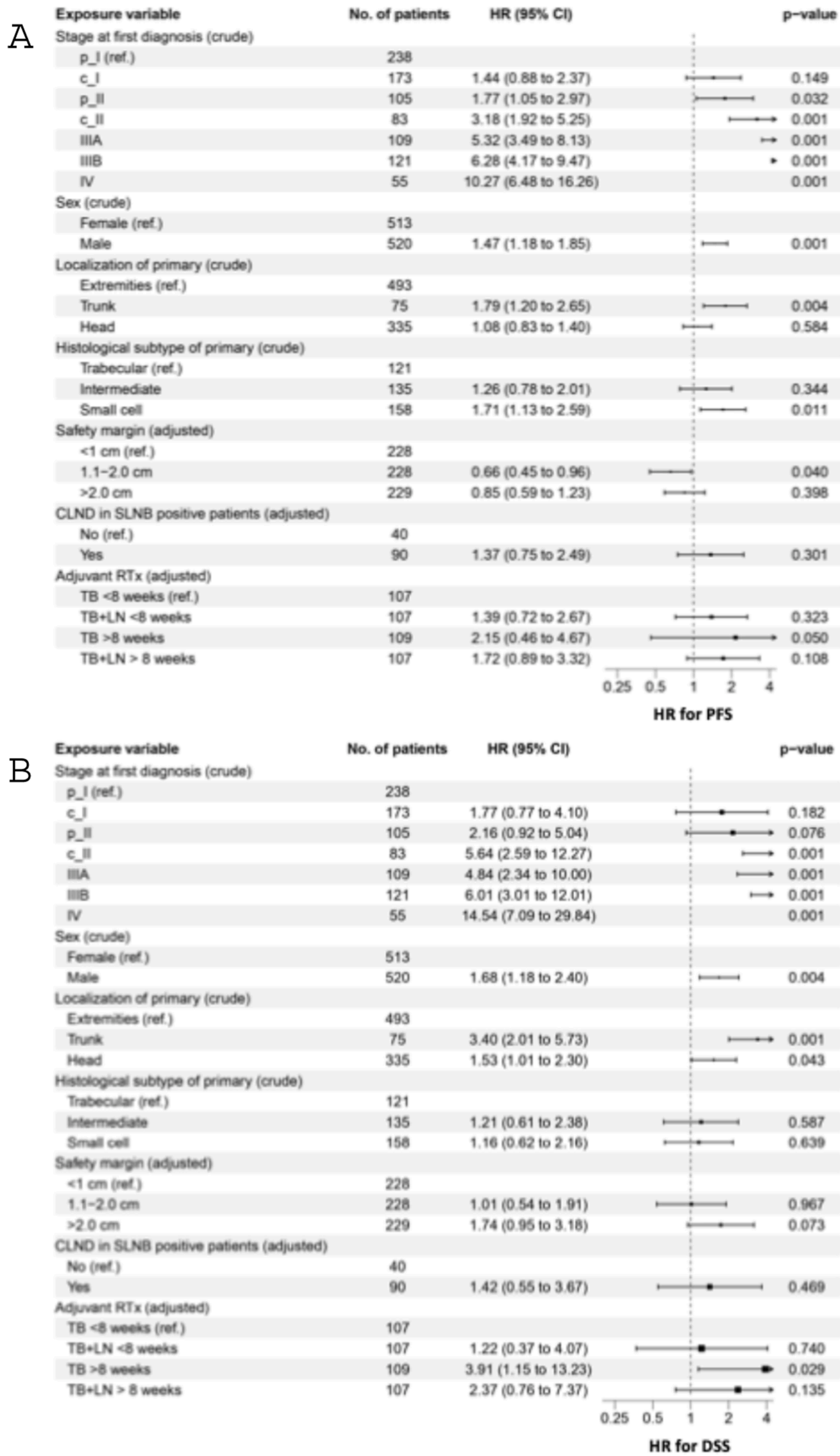

Figure 2. Hazard ratio for progression-free probability (A) and disease-specific (B) survival with the corresponding 95 % confidence intervals were estimated using the Cox proportional hazards model. If indicated to estimate adjusted hazard ratios, we used inverse probability treatment weighting with weights derived from age, sex and tumor stage.

surgical safety margins on disease progression and disease-specific mortality was less pronounced than anticipated. A safety margin of 1–2 cm lead to a significantly improved PFP (HR adjusted for age, sex, and stage 0.66; 95 %CI 0.45–0.98; Figure 3A). DSS improved to a lesser degree (HR adjusted for age, sex, and stage 0.58; 95 % CI 0.33–1.03; Figure 3B), Similar results were detected after further adjustment for adjuvant radiotherapy (Supplementary Figure S3). Importantly, safety margins exceeding 2 cm did not confer any additional benefit in terms of survival outcomes. Tumor stage-specific analyses of safety margins showed similar results (Supplementary Table 6 and 7). A competing risk model showed higher cumulative incidence for progression or MCC-related death in patients with safety margin less than 1 cm compared to patients with safety margin of 1–2 cm and greater than 2 cm (Figure 4A). However, the curves remain very close to each other.

According to national and international guidelines, CLND and/or radiotherapy is recommended in patients with micrometastatic disease in the sentinel lymph nodes.[4–7] In our cohort, of patients with known CLND status 90 of 130 patients (69.2 %) with a positive SLNB underwent CLND. Notably, our results show that CLND did not result in a significant improvement in disease outcomes in patients with pathologically confirmed micrometastases, indicating the limited prognostic benefit of this intervention. Interestingly, adjusted hazard ratios for PFP and DSS were unexpectedly even lower in patients who did not undergo CLND (PFP HR 0.73, 95 %CI 0.40–1.33; DSS HR 0.70, 95 %CI 0.27–1.82; both adjusted for age, sex, and disease stage; Figure 3C and D). At 60 months, PFP and DSS probabilities were 59.8 % (95 %CI 42.4–84.3) and 79.1 % (95 %CI 64.0–97.9), respectively, for patients without CLND, compared with 36.9 % (95 %CI 26.8–50.9) and 70.3 % (95 %CI 59.1–83.5) for those who underwent CLND.

Therapeutic recommendations for stage I–III MCC patients include adjuvant post-surgical radiotherapy to the tumor bed and/or regional lymph node area in cases of nodal involvement. The observations in our cohort not only confirm the benefit of adjuvant radiotherapy, but also show that the timing of radiotherapy significantly influences clinical outcomes (Figure 3E and F). Applying adjuvant radiotherapy more than 8 weeks post-diagnosis were associated with poorer PFP and DSS with a HR adjusted to age, sex and disease stage of 1.17 (95 %CI 0.78–1.76) for PFP and 1.82 (95 %CI 0.91–3.64) for DSS (Figure 3E and F). A competing risk model showed higher cumulative incidence for progression or MCC-related death in patients with radiotherapy later than 8 weeks compared to patients with radiotherapy within 8 weeks after first diagnosis (Figure 4B). Stage-specific analyses of time of radiotherapy showed similar results (Supplementary Table 8 and 9). A critical finding was the limited benefit observed in patients with stages cII, pII and IIIA, when radiotherapy was extended from the tumor bed to the regional lymph node basin. Indeed, additional radiation of the lymph node region did not improve outcomes, with an adjusted HR of 1.73 (95 %CI 0.84–3.57) for PFP and 119 (95 %CI 0.44–3.19) for DSS; Figure 3G and H). Multivariable Cox regression confirmed this observation (Supplementary Table 10). These findings underscore the nuanced interplay between timing and extent of radiotherapy, emphasizing the potential importance of tailored treatment strategies to optimize disease control in MCC patients.

Discussion

While the incidence and survival data of a rare cancer such as MCC can be most reliably obtained from cancer registries covering large populations, these primarily assess the probability of survival at the population level.[2,3,16] In contrast, specialized disease registries offer detailed insights into patient and tumor characteristics, therapeutic management, and disease progression. Therefore, we utilized our large national database comprising 1049 prospectively enrolled MCC patients with a median follow-up exceeding 5 years to evaluate not only known risk factors, but also the impact of locoregional therapy. Notably, this dataset is independent of the extensively studied SEER registry.[28–30] Possible bias stemming from the predominant reporting by referral centers to this registry was excluded by comparison of baseline patient and tumor characteristics with data from the largest German population-based cancer registry.[16] This notion was further substantiated by the fact that we clearly confirmed previously reported factors affecting the risk of recurrence, such as the stage at diagnosis, sex, age, site of the primary tumor, and immunosuppression.[2,10,28,31] In addition, our extensive clinical dataset allowed us to assess the prognostic significance of controversial factors, including the histological subtype of the primary tumor. Evidence supporting the clinical relevance of histologic classifications remains limited, leading to ongoing discussion and occasional dismissal.[32] Our analysis aligns with the widely accepted view that small-cell MCCs have a higher recurrence risk than trabecular subtypes. However, this observation still requires confirmation through a study with a centralized pathological review, such as the ongoing Tissue Registry in MCC (MCC TRIM; EMA EUPAS25338). Crucially, we are the first to demonstrate that multifocal metastasis at initial recurrence predicts a high likelihood of further metastatic spread and increased case fatality. This finding likely reflects biological mechanisms such as genetic instability and tumor heterogeneity.[33] Notably, multifocal metastases, whether locoregional or systemic, were a stronger adverse prognostic factor than unilocular systemic metastases.

Initial MCC treatment typically involves stage-dependent multimodal approaches. [4–7] German national guidelines recommend wide local excision with margins up to 3 cm and SLNB.[6] Postoperative radiation to the tumor bed and draining lymph nodes is frequently used. In case of a positive SLNB, CLND or lymph node irradiation, either alone or combined, are advised. However, these recommendations largely rely on empirical evidence, anecdotal reports, or small case series, leaving significant uncertainty about their clinical impact. Our comprehensive dataset, encompassing detailed records of interventions and outcomes in a sizeable cohort, enabled a robust evaluation of the practical benefits of the recommended locoregional treatments. The findings reveal that the impact of these interventions on disease progression often falls short of expectations, highlighting the need for a critical reassessment of certain therapeutic recommendations. We consciously restricted our analysis to the period before the widespread adoption of immune checkpoint inhibitors for adjuvant or therapeutic MCC treatment, ensuring a focus on the natural course of the disease, largely unaffected by effective systemic therapies.[14,15]

Our primary finding was that the influence of surgical safety margins on disease progression and disease-specific survival (DSS) was less pronounced than anticipated. Progression-free probability (PFP) improved with margins of 1–2 cm across all analyses, whereas the association with DSS was less evident. Given prior findings by Tarabadkar et al.[34], suggesting that narrow margins (<1 cm) combined with adjuvant radiotherapy may achieve disease control comparable to wider margins, we conducted PFP and DSS analyses adjusting for age, sex, tumor stage, and adjuvant radiotherapy (Supplementary Figure S3). This adjusted analysis confirmed that margins of 1–2 cm were associated with superior PFP and DSS compared with narrower margins. Notably, however, margins exceeding 2 cm conferred no additional benefit in any analysis. This is in line with prior study results.[35] Furthermore, performing a CLND in patients with micrometastases identified through pathological examination of the SLNB did not result in a clinically meaningful improvement. While a prospective randomized study would undeniably represent the gold standard in determining the most effective therapeutic approach, the feasibility of conducting such a trial for a rare malignancy like MCC remains highly improbable. Nevertheless, the current evidence suggests that a cautious, yet pragmatic approach, grounded in available clinical data and expert consensus, may offer the most realistic path forward for patient management.

For stage I–III MCC, guidelines recommend adjuvant radiotherapy to the tumor bed and/or regional lymph nodes in cases of nodal involvement. Emerging evidence highlights the importance of timely initiation of readiotherapy, with reports suggesting optimal outcomes when

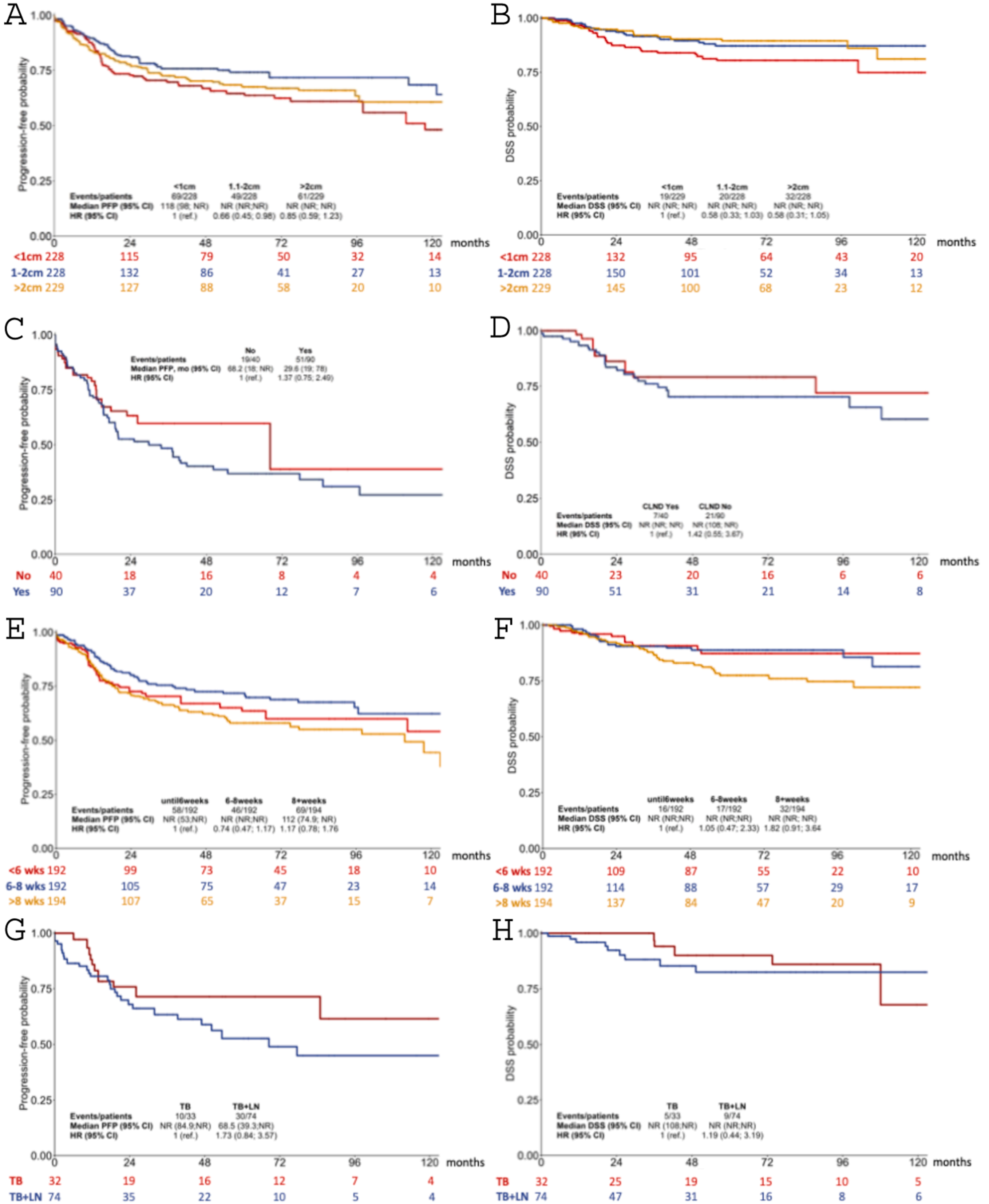

Figure 3. Progression-free probability and disease-specific survival by excision margin (A, B), complete lymph node dissection in SLNB positive patients (C, D) and time of adjuvant radiotherapy (E, F) as well as extension of the radiation field (G,H). The Kaplan-Meier estimates were calculated after adjustment for age, sex, and tumor stage using inverse probability treatment weighting. For the impact of extensions extension of the radiation field we focused on patients in stages cII, pII and IIIA (G,H).

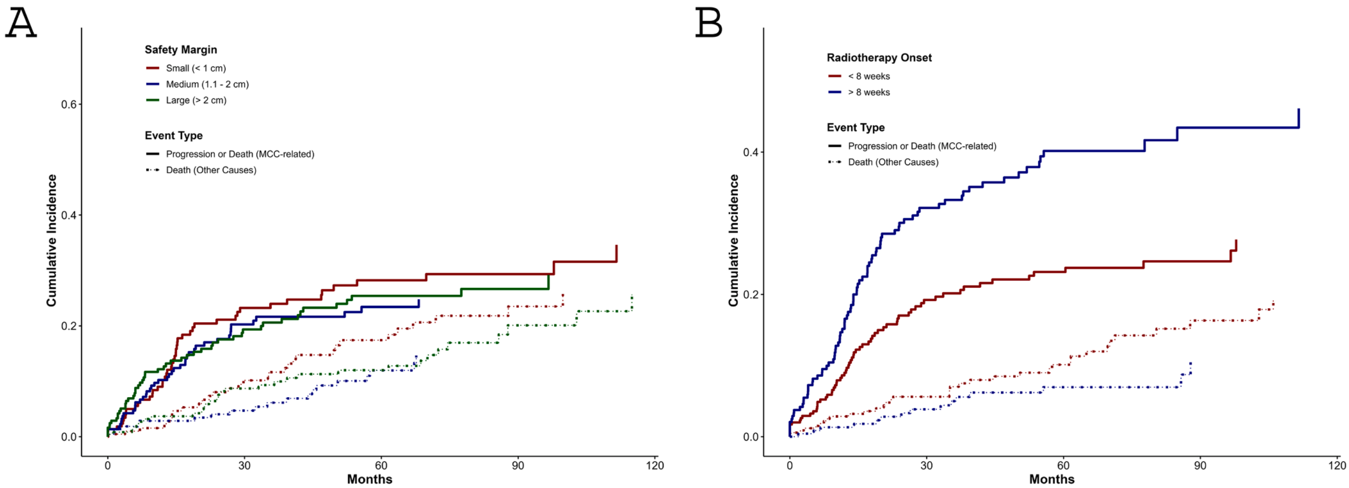

Figure 4. Cumulative Incidence Functions for competing risks. This figure shows the cumulative incidence of two competing events—tumor recurrence (solid lines) and death without tumor recurrence (dashed lines)—over time by excision margin (A) and time of adjuvant radiotherapy (B). Data were analyzed using Fine and Gray competing risks regression.

started within 8–10 weeks of tumor excision.[36,37] Our findings confirm that the timing of adjuvant radiotherapy critically affects clinical outcomes, with starting it more than 8 weeks after diagnosis significantly compromising disease-specific survival (DSS). Conversely, extending timely radiotherapy from the tumor bed to the lymph node basin offered minimal to no additional benefit, consistent with recent findings by Pairawan et al.[9] This notion was further confirmed by multivariable analysis (Supplementary. Table 7). These observations may reflect the immunobiology of MCC, as surgical resection or irradiation of locoregional tumor-draining lymph nodes can negatively impact anti-tumor immune responses.[38] Specifically, these interventions may remove or damage MCC-specific lymphocytes, disrupt antigen presentation thus impairing the development and maintenance of effective anti-tumor immunity, thereby influencing the subsequent clinical trajectory of the disease.[39]

Comprehensive data on first progression allowed us to examine its characteristics and subsequent clinical trajectory in detail. Nearly three-quarters of events occurred within the first two years after diagnosis. While most recurrences were locoregional, almost half of patients developed distant metastases, either with or independent of locoregional disease. Notably, most distant metastases within the first two years should be detectable on clinical and ultrasound examination, affecting the skin, lymph nodes, and soft tissue. In contrast, organ metastases appeared later but were associated with significantly reduced disease-specific survival. These findings raise questions about the current follow-up protocol for MCC and suggest that imaging of internal organs should be maintained or even intensified beyond two years post-diagnosis.[4–7]

Our analysis also highlighted notable limitations in adherence to national guidelines for the management of MCC patients within German-speaking countries. This discrepancy can, in part, be attributed to the paucity of robust evidence supporting the efficacy of the recommended interventions. The limited data available on the potential benefits of these treatments may contribute to hesitancy in their widespread implementation, underscoring the need for further research to solidify clinical recommendations.[6] For instance, SLNB was performed in only 60 % of patients. This is concerning, as our data confirm that pathological staging provides significantly better prognostic accuracy than clinical staging alone.

Limitations of our study include the fact that not all clinical and therapeutic details are documented for all patients. In particular, the histopathological characteristics that have already been proposed as prognostic biomarkers in smaller case series are missing for many patients (likely due to it's controversial impact on prognostication). In addition, there was no central confirmation of the histopathological diagnosis, which can certainly lead to bias in an entity such as MCC that is not always trivial to diagnose. To address the latter problems, we have initiated the Tissue Registry in MCC (MCC TRIM; EMA EUPAS25338) in which in addition to detailed clinical data. tumor tissue is collected, thus enabling a central investigation of tumor tissues. A further limitation are missing data in variables such as safety margins and adjuvant radiotherapy, which we tried to address by multiple imputation. Still, missing data must be taken into consideration when interpreting the results.

The circumstance that our analysis of 1049 MCC patients from the pre-ICI era confirms previously identified risk factors for disease recurrence and MCC-specific death, underscores the clinical relevance of our findings regarding currently recommended locoregional therapeutic interventions. Specifically, surgical margins greater than 2 cm, adjuvant radiotherapy extending beyond the tumor bed (when administered within 8 weeks of primary diagnosis), and a CLND in SLNB-positive patients offer limited, if any, benefit, particularly in terms of disease-specific survival. These results suggest that the scope of recommended locoregional treatments for MCC should be critically re-evaluated.

Contributions

GCL, SU and JCB contributed to the conception and design of the study, data acquisition, data interpretation, manuscript writing, and finalization of the report. UL, AG, TE, AH, CP, TG, RH, FM, JCH, FM, PM, PT, GN, and MH contributed to patients' recruitment, data acquisition, and manuscript review. JCB was responsible for project supervision, resource allocation, and funding acquisition. TG oversaw the data management operations. AS and WG are the statistician and epidemiologists responsible for the statistical analysis plan and the biometrical analyses. GCL, SU and JCB directly accessed and verified the underlying data. GCL and MD were responsible for the visualization of the results. SU and JCB were responsible for the decision to submit the manuscript for publication. All authors reviewed and approved the final version of the manuscript. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Disclosure

GL has received travel support from Sun Pharma. TG has received speakers and/or advisory board honoraria from BMS, Sanofi-Genzyme, MSD, Novartis Pharma, Roche, Abbvie, Almirall, Janssen, Lilly, Pfizer, Pierre Fabre, Merck-Serono, outside the submitted work. RH is employee of Helios Kliniken Erfurt GmbH. FM has received honoraria from Novartis, Bristol-Myers Squibb, Merck Sharp & Dohme, Pierre Fabre, Sanofi Genzyme, Sun Pharma and travel support from Novartis, Sun Pharma, Roche, Pierre Fabre and Merck Sharp & Dohme, outside the submitted work. JCH declares research support from Bristol Myers Squibb, Sanofi and Sunpharma; speakers honoraria from Amgen, Bristol Myers Squibb, Delcath, GSK, Immunocore, MSD, Novartis, Pierre Fabre, Sanofi and Sunpharma and advisory board honoraria from GSK, Pierre Fabre and Sunpharma, and travel support from Bristol Myers Squibb, Iovance and Sunpharma. PT served as consultant and/or received honoraria form Almirall, Bristol Myers Squibb, Biofrontera, Kyowa Kirin, L'Orèal, Merck Sharp & Dohme, Novartis, Pierre-Fabre, Sanofi, 4SC, and travel support from Bristol Myers Squibb outside the submitted work. SU declares research support from Bristol Myers Squibb and Merck Serono; speakers and advisory board honoraria from Bristol Myers Squibb, Merck Sharp & Dohme, Merck Serono, Novartis and Roche, and travel support from Bristol Myers Squibb, Merck Sharp & Dohme, and Pierre Fabre. JCB is receiving speaker's bureau honoraria from Amgen, MerckSerono, Pfizer, Sanofi and Sun Pharma and is a paid consultant/advisory board member/DSMB member for Almirall, Boehringer Ingelheim, ICON, Pfizer, Regeneron and Sanofi. His group receives research grants from Alcedis, Merck Serono/IQVIA, and Regeneron. All other authors declare no competing interests.

Data sharing

The datasets generated during and/or analyzed during the current study are available for qualified researchers from the corresponding authors upon request.

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Galetzka Wolfgang: Writing – review & editing, Supervision, Methodology. **Ugurel Selma:** Writing – review & editing, Writing – original draft, Supervision, Data curation, Conceptualization. **Herbst Rudolf:** Writing – review & editing, Resources, Data curation. **Meiß Frank:** Writing – review & editing, Resources, Data curation. **Mohr Peter:** Writing – review & editing, Resources, Data curation. **Lodde Georg:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Becker Jurgen C:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization. **Meier Friedegund:** Writing – review & editing, Resources, Data curation. **Hassel Jessica C.:** Writing – review & editing, Resources, Data curation. **Hecht Markus:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Eigentler Thomas:** Writing – review & editing, Resources, Data curation. **Stang Andreas:** Writing – review & editing, Supervision, Methodology, Formal analysis. **Hauschild Axel:** Writing – review & editing, Resources, Data curation, Conceptualization. **Terheyden Patrick:** Writing – review & editing, Resources, Data curation. **Leiter Ulrike:** Writing – review & editing, Resources, Data curation. **Nikolakis Georgios:** Writing – review & editing, Resources, Data curation. **Gesierich Anja:** Writing – review & editing, Resources, Data curation. **Dalkoohi Mazdak:** Writing – review & editing, Visualization, Software, Methodology. **Pföhler Claudia:** Writing – review & editing, Resources, Data curation. **Gambichler Thilo:** Writing – review & editing, Resources, Data curation.

Declaration of Competing Interest

GL has received travel support from Sun Pharma. TG has received speakers and/or advisory board honoraria from BMS, Sanofi-Genzyme, MSD, Novartis Pharma, Roche, Abbvie, Almirall, Janssen, Lilly, Pfizer, Pierre Fabre, Merck-Serono, outside the submitted work. RH is employee of Helios Kliniken Erfurt GmbH. FM has received honoraria from Novartis, Bristol-Myers Squibb, Merck Sharp & Dohme, Pierre Fabre, Sanofi Genzyme, Sun Pharma and travel support from Novartis, Sun Pharma, Roche, Pierre Fabre and Merck Sharp & Dohme, outside the submitted work. JCH declares research support from Bristol Myers Squibb, Sanofi and Sunpharma; speakers honoraria from Amgen, Bristol Myers Squibb, Delcath, GSK, Immunocore, MSD, Novartis, Pierre Fabre, Sanofi and Sunpharma and advisory board honoraria from GSK, Pierre Fabre and Sunpharma, and travel support from Bristol Myers Squibb, Iovance and Sunpharma. PT served as consultant and/or received honoraria form Almirall, Bristol Myers Squibb, Biofrontera, Kyowa Kirin, L'Orèal, Merck Sharp & Dohme, Novartis, Pierre-Fabre, Sanofi, 4SC, and travel support from Bristol Myers Squibb outside the submitted work. SU declares research support from Bristol Myers Squibb and Merck Serono; speakers and advisory board honoraria from Bristol Myers Squibb, Merck Sharp & Dohme, Merck Serono, Novartis and Roche, and travel support from Bristol Myers Squibb, Merck Sharp & Dohme, and Pierre Fabre. JCB is receiving speaker's bureau honoraria from Amgen, MerckSerono, Pfizer, Sanofi and Sun Pharma and is a paid consultant/advisory board member/DSMB member for Almirall, Boehringer Ingelheim, ICON, Pfizer, Regeneron and Sanofi. His group receives research grants from Alcedis, Merck Serono/IQVIA, and Regeneron. All other authors declare no competing interests.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ejca.2025.115406.

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